

Precept 3 Outline

ECO 202 TAs

September 24, 2026

1 Exam Prep (if read before exam)

1. Existing Practice Material:

- (a) Each lecture note contains short practice questions and checkpoints
- (b) Each lecture also has practice problems associated with the lecture, with solutions
- (c) For each precept we post notes with some additional problems not covered in precept, with solutions
- (d) Two practice exams posted with solutions on the github!

2. Other Logistics:

- No calculators allowed on exam!
- No coding/AI problems on exam
- Material up to and including Monday, September 21 lecture is fair game

2 Precept 3 (Lecture 4, 5) Core Concepts

2.1 Lecture 4

1. Correlation, covariance, and relationship between them
2. Least squares minimization problem
 - How they lead to formulas for b_1, b_0
 - Residuals and properties of residuals (Note: can talk through what residuals look like on graph)
3. Properties of Least squares
 - Fitted line passes through $\bar{x}, \bar{y}$
4. R^2 and r (R squared versus correlation, and relationship between them)

2.2 Lecture 5

1. Potential outcome notation: D_i , $Y_i(1)$, and $Y_i(0)$
2. Average Treatment Effect (ATE)

3 Precept practice problems

- Let y_i , $i = 1, \dots, 265$, be the fraction of students in school district i who qualify for a subsidized lunch, and let x_i be average household income in that school district, measured in thousands of dollars. Suppose

$$\bar{x} = 38.0, \quad s_x = 14.0, \quad \bar{y} = 0.26, \quad s_y = 0.18, \quad r = -0.73.$$

- Given these data, what is your best guess of the fraction of students who qualify for subsidized lunch in a district with average household income equal to \$24,000 (so $x = 24.0$)? Explain.
 - Given these data, what is your best guess of average household income in a school district where $0.45 = 45\%$ of students qualify for subsidized lunch? Explain.
 - Why is it problematic to use linear regression here to predict the fraction of students receiving subsidized lunches in very wealthy school districts?
- In an effort to understand the effects of the Atkins diet, a randomized experiment is run with 320 initially overweight mice. All mice receive food with the same total number of calories, but the calorie intake is achieved with five different levels of carbohydrates. Let x denote carbohydrate intake in grams per day. After 28 days, the mice are weighed; let y denote weight in grams. We know that

$$\bar{x} = 16.2, \quad \bar{y} = 167.3, \quad s_x = 6.5, \quad s_y = 35.8,$$

and the slope of the least-squares regression line is $b = 4.6$.

- Why can one be reasonably confident that this evidence points toward a causal relationship between diet and weight in mice? Compute a relevant measure of fit and explain why a causal interpretation is appropriate using relevant terminology.
 - Suppose x were instead measured as total carbohydrate intake, in grams, over the 28 days of the experiment. What would be the new regression line, including its intercept and slope?
- (Lecture 5 Practice, Problem 2)** The table displays both potential outcomes only so the causal quantities can be checked; real data would reveal at most one potential outcome per unit.

Unit	$Y_i(0)$	$Y_i(1)$	D_i	Observed outcome (Y_i)
A	4	7	1	7
B	6	7	1	7
C	2	4	0	2
D	5	6	0	5

- Calculate every individual treatment effect and the average treatment effect (ATE) across the four units.
- Verify the observed-outcome identity

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

for one treated and one untreated unit. Identify which potential outcome would be missing for each unit in an actual study.

- Calculate the average treatment effect among the treated units.
- Calculate the observed treated-minus-untreated mean difference.
- Calculate the untreated-potential-outcome selection term $\overline{Y(0)}_{D=1} - \overline{Y(0)}_{D=0}$, and verify that the average treatment effect among the treated plus this term equals the observed difference.
- Explain in words why the observed difference is not the ATE and why the counterfactual problem would prevent this decomposition from being calculated directly in real observational data.

4. The administration of Princeton University notices a strong positive correlation between students' academic achievement and whether they play an instrument. Admission Officer A hears of this finding and says, "Given that we want to admit students who do well academically, we should admit more students who play an instrument." Admission Officer B counters, "But this is just a correlation, and it does not imply that playing an instrument causes students to do well in their studies at Princeton University."

Who is right? Explain using relevant terminology, and relate your discussion to the specific issue of playing an instrument and academic success.

4 Extra practice problems

1. Using similar techniques as for temperature, researchers reconstructed the average concentration of CO₂ in Earth's atmosphere over the last 410,000 years, measured in parts per million by volume. The variable x_i , $i = 1, \dots, 82$, contains the historic concentrations of CO₂ in 5,000-year intervals. The variable y_i records the difference between today's average global temperature and the historic global temperature, in degrees Celsius.

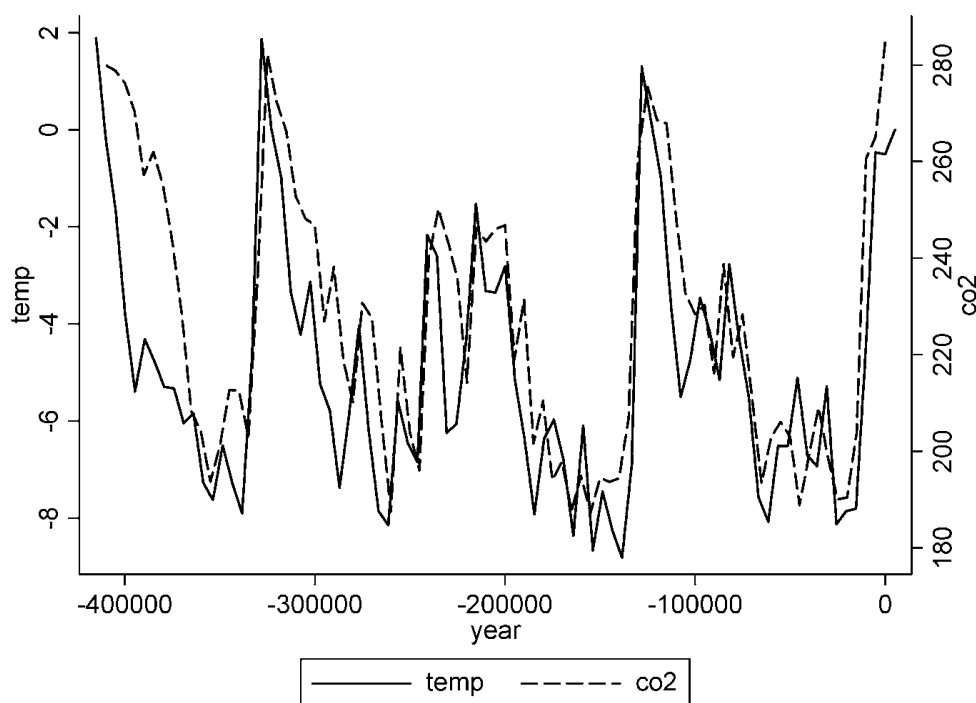
The means and standard deviations are

$$\bar{y} = -4.9, \quad \bar{x} = 227.3, \quad s_y = 2.64, \quad s_x = 28.3$$

The least-squares regression line for a regression of y on x is

$$\hat{y} = 0.081x - 23.31$$

- (a) What is your prediction of the historic relative temperature, in degrees Celsius, for a time when the CO₂ concentration was one standard deviation below its overall average?
- (b) How confident are you about your prediction in part (a)? Compute a relevant measure and describe its interpretation for assessing predictions based on regression lines.
- (c) The CO₂ concentration was predicted to be 570 parts per million by volume in the year 2050. Compute the z -score of this value of x . Would you have much confidence in a prediction of the relative temperature in 2050 based on the regression line, given your result?
- (d) The figure below plots the joint evolution of relative temperature and CO₂ concentration over the last 410,000 years. Does the plot suggest that declines in CO₂ caused temperature to decline, or rather that declines in temperature caused CO₂ to decline? Why?



Historic relative temperature (solid) and CO₂ concentration (dashed).

2. (a) Let $\hat{y}_i$, $i = 1, \dots, n$, be the predicted values from a regression of y_i on x_i . Suppose

$$\bar{x} = 3.4, \quad \bar{y} = -5.9, \quad \bar{\hat{y}} = -5.9, \quad s_x = 3.4, \quad s_y = 6.1, \quad s_{\hat{y}} = 2.2.$$

Say as much as you can about the correlation between x_i and y_i . Explain.

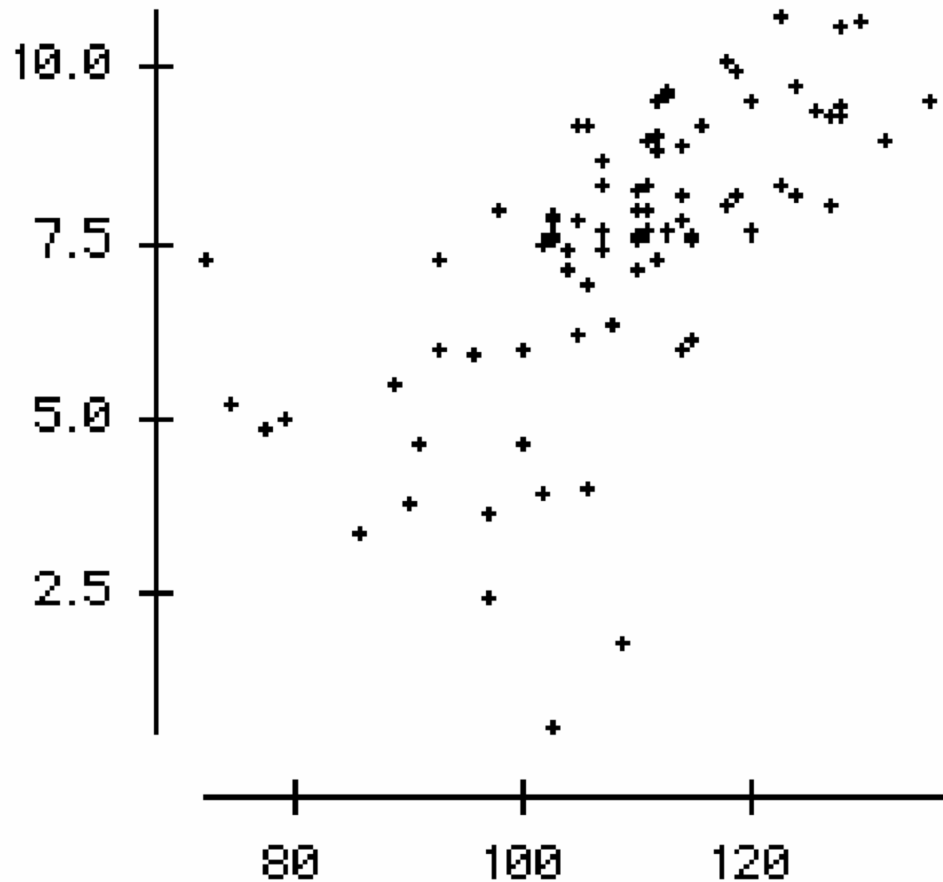
- (b) Give an example of an association between two variables that is plausibly due to a common response.
3. State whether each claim is true or false and explain briefly.
- (a) While the mean and standard deviation are not resistant measures, the regression slope is resistant.
 - (b) If there is no linear association between two variables, then the correlation r is close to zero.
 - (c) The mean of the regression residuals is always zero.
 - (d) If x and y are standardized, then the regression line from regressing y on x coincides with the regression line from regressing x on y .
 - (e) One explanation for an observed association between two variables is that one causes the other.
4. (a) The height of men in the United States is approximately normally distributed with mean 68 inches and standard deviation 5 inches. Their weight has mean 168 pounds and standard deviation 25 pounds. The correlation between weight and height is 0.35. What is your best guess of a man's height if you are told that he weighs 160 pounds?
- (b) Identify two mistakes in the following data description:
 "The level of snow y_i , $i = 1, \dots, 365$, on Mount McKinley, measured daily in inches, has mean 83 inches, median 78 inches, interquartile range 42 inches, and variance 900 inches². The distribution

is skewed to the right. Temperature x_i , in degrees Fahrenheit, has mean 18 and standard deviation 20. The least-squares regression line

$$\hat{y} = a + bx$$

has $a = 47$ degrees Fahrenheit and $b = 2$ inches per degree Fahrenheit.”

- (c) True or false: the correlation of the data in the supplied scatterplot is closer to 0.2 than to 0.6.



Scatterplot supplied with Midterm I 2006, Question 4(b).

- (d) True or false: outliers of the explanatory variable are often influential in determining the least-squares regression line.